

2017 F=ma Exam: Problem 21

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We are given

$$f = \frac{Mv_2}{mv_0}$$

where v_2 is the velocity of M after the collision.



For an elastic collision, we have

$$v_2 = \frac{2m}{m+M}v_0$$

Then

$$f = \frac{M}{mv_0} \frac{2mv_0}{m+M} = \frac{2M}{m+M} = \frac{2}{1+r}$$

where $r = m/M$. The maximum possible f occurs when $r \rightarrow 0$ i.e. $M \rightarrow \infty$ so

$$f_{\max} = \frac{2}{1+0} = 2$$

Thus, the answer is D.